

CRC RevEng

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Catalogue of parametrised CRC algorithms with 16 bits



CRC-16/ARC

```
width=16 poly=0x8005 init=0x0000 refin=true refout=true  
xorout=0x0000 check=0xbb3d residue=0x0000 name="CRC-16/ARC"
```

- Class: attested
- Alias: **ARC**, **CRC-16**, **CRC-16/LHA**, **CRC-IBM**
- AUTOSAR (24 November 2022), [AUTOSAR Classic Platform](#) release R22-11, [Specification of CRC Routines](#)
 - I Comprehensive primer on CRC theory (Section 7.1, pp.19–22)
 - I All parameters (Section 7.2.2.2, p.27)
 - IV 7 codewords (Section 7.2.2.2, p.27)
 - 000000000000
 - F20183E1C2
 - 0FAA0055E30B
 - 00FF5511CF6C
 - 332255AABBCCDDEEFF98AE
 - 926B554EE2
 - FFFFFFFF0194
- System Enhancement Associates (24 October 1986), ARC 5.20
 - II Implementation
- Haruyasu Yoshizaki (10 January 1996), LHA 2.55E
 - II Implementation
- Rahul Dhesi (19 April 1996), ZOO 2.1a
 - II Implementation
- Lammert Bies (August 2011), [CRC calculator](#)
 - II Implementation
- PVL Team (25 October 2008), CRC .NET control, version 14.0.0.0
 - II Implementation (CRC16_arc)
- Dr Ross N. Williams (19 August 1993), ["A Painless Guide to CRC Error Detection Algorithms"](#) (courtesy of the Internet Archive)
 - III All parameters (except Residue)
- Emil Lenchak, Texas Instruments, Inc. (June 2018), [CRC Implementation With MSP430](#)
 - III All parameters (except Residue) (Section 4.6, p.6)
- Altera Corporation (April 1999), [crc MegaCore Function Data Sheet](#), version 2 (courtesy of the Internet Archive)
 - III All parameters (except Residue) (p.6)
- Unique effective solution of codeword set
- Created: 30 March 2005
- Updated: 2 August 2023

CRC-16/CDMA2000

width=16 poly=0xc867 init=0xffff refin=false refout=false xorout=0x0000 check=0x4c06
residue=0x0000 name="CRC-16/CDMA2000"

- Class: **academic**
- 3rd Generation Partnership Project 2 (3GPP2) (September 2009), [Physical layer standard for cdma2000 spread spectrum systems](#), revision E, version 1.0 (courtesy of the Internet Archive)
 - I Definition: Width, Poly (Section 2.1.3.1.4.1.1, p.2-95)
 - I Full description (Section 2.1.3.1.4.1, p.2-95)
 - I Shift register diagram (Figure 2.1.3.1.4.1.1-1, p.2-95)
- Agilent Technologies, Advanced Design System 2008 Update 2, [CDMA2K_CRC_Coder module documentation](#) (courtesy of the Internet Archive)
 - III Full description
- Created: 14 October 2013
- Updated: 28 July 2023

CRC-16/CMS

width=16 poly=0x8005 init=0xffff refin=false refout=false xorout=0x0000 check=0xae7
residue=0x0000 name="CRC-16/CMS"

- Class: **third-party**
- Dainius Šimelevičius, Remi K. Mommsen *et al.* (16 December 2016), CERN CMS Online Software (XDAQ) release 14, Event Builder source package, files [daq/evb/include/evb/CRCCalculator.h](#), [daq/evb/src/common/CRCCalculator.cc](#)
 - III All parameters (except Residue)
 - III Code: C++
- "movilstore" (24 January 2011), Computer Interfacing Forum [topic 1650](#) (courtesy of the Internet Archive)
 - III Reference to application of algorithm on Samsung mobiles
 - IV 5 codewords
 - 0200080078110000F00F0000F7E0
 - 020008005B110000F00F00004725
 - 0200080050110000F00F0000F71F
 - 0200080024110000F00F00003636
 - 02000800A6100000F00F0000141A
- "xtrusion" (28 November 2010), Computer Interfacing Forum [topic 1627](#) (courtesy of the Internet Archive)
 - IV 12 codewords
 - 0200080034100000B0090000B725
 - 0200080029100000B0090000E76E
 - 0200080002100000B0090000D79B
 - 0200080005100000B0090000A789
 - 0200080078100000B0090000F696
 - 00050900000000003030303000D55E
 - 01000400051000003793
 - 0200080034100000F00F00003740
 - 0200080029100000F00F0000670B
 - 0200080002100000F00F000057FE
 - 0200080005100000F00F000027EC
 - 0200080078100000F00F000076F3
- "Robert" ("xtrusion") (25 December 2010), Computer Interfacing Forum [topic 1636](#) (courtesy of the Internet Archive)
 - IV 5 codewords

- 02000C00301000009A0100000000000837B
- 02000C00301000009A0100009A0100004B55
- 02000C00301000009A01000034030000934E
- 02000C00301000009A010000CE0400005B09
- 02000C00301000009A01000068060000A311
- Unique effective solution of codeword set
- Created: 13 July 2016
- Updated: 2 August 2023

CRC-16/DDS-110

width=16 poly=0x8005 init=0x800d refin=false refout=false xorout=0x0000 check=0x9ecf
residue=0x0000 name="CRC-16/DDS-110"

- Class: attested
- Used in the ELV DDS 110 function generator. Init value is equivalent to an augment of 0xFFFF prepended to the message.
- ELV Elektronik AG (March 2008), [Software-Schnittstelle der Funktionsgeneratoren DDS 10/DDS 110](#) (in German; courtesy of the Internet Archive)
 - I Definition: Width, Poly, CRC byte order (p.67)
 - IV 1 codeword (escaped according to the text) (p.67)
 - 0200108200731082F7FE (source: 02000200731082FEF7)
- "wortel", "Gammatester" (19 July 2009), Computer Interfacing Forum [topic 1372](#) (courtesy of the Internet Archive)
 - III All parameters (except Check, Residue) ("Gammatester")
 - IV 3 codewords ("wortel")
 - 02001082007800ED96
 - 020010820078016D93
 - 020005006600057A586435
- Unique effective solution of codeword set
- Created: 30 November 2009
- Updated: 28 December 2019

CRC-16/DECT-R

width=16 poly=0x0589 init=0x0000 refin=false refout=false xorout=0x0001 check=0x007e
residue=0x0589 name="CRC-16/DECT-R"

- Class: attested
- Alias: **R-CRC-16**
- Used in DECT A-fields.
- ETSI [EN 300 175-3](#) version 2.5.1 (August 2013)
 - I Full mathematical description (Section 6.2.5.2, p.99)
 - I Performance of polynomial (Annex B, p.297)
- Thomas Pircher (5 November 2022), [pycrc 0.10.0](#)
 - II Implementation
- Pavel Yazev (9 September 2015), gr-dect2 module [gr-dect2/lib/packet_decoder_impl.cc](#)
 - III Code: C++
- Created: 7 December 2009
- Updated: 3 August 2023

CRC-16/DECT-X

- Patrick Seem, Texas Instruments, Inc. (23 October 2008), [AN067: Wireless MBUS Implementation with CC1101 and MSP430](#)
 - **III** Definition: Width, Poly, Init, XorOut (Section 5.4, p.9)
 - **III** Describes synchronous transfer with MSBs sent first, implying RefIn and RefOut (Sections 7.2.1, 7.3.1, pp.13–14)
- Dr.-Ing. Thomas Weinzierl, Weinzierl Engineering GmbH (July 2005), [Stack Implementation for KNX-RF](#) (courtesy of the Internet Archive)
 - **III** Definition: Poly (Section 2.3, p.3)
 - **III** CRC byte order, implying RefIn and RefOut (Section 2.3, p.2, etc.)
 - Radio link corresponds to Link A in AN067 (Section 2.2, p.2 vs. AN067 table 3, p.6)
- "Tim" (6 June 2004), Control.com [forum post](#)
 - **III** Width, Poly cited for ISO/IEC 60870-5-2
- "MarcPo", "Gammatester" (30 September 2008), Computer Interfacing Forum [topic 925](#) (courtesy of the Internet Archive)
 - **III** All parameters (except Check, Residue) ("Gammatester")
 - **IV** 1 codeword ("MarcPo")
 - 780B134365871E6D
- "Chris P." (27 April 2009), Computer Interfacing Forum [topic 1315](#) (courtesy of the Internet Archive)
 - **IV** 1 codeword
 - 07400001AAAA0000363E
- Unique effective solution of codeword set
- Created: 1 October 2008
- Updated: 28 December 2019

CRC-16/GENIBUS

width=16 poly=0x1021 init=0xffff refin=false refout=false xorout=0xffff check=0xd64e
residue=0x1d0f name="CRC-16/GENIBUS"

- Class: attested
- Alias: **CRC-16/DARC**, **CRC-16/EPC**, **CRC-16/EPC-C1G2**, **CRC-16/I-CODE**
- Used in standardised RFID tags. Presented high byte first.
- EPCglobal Inc™ (4 September 2018), [EPC UHF Gen2 Air Interface Protocol: EPC™ Radio-Frequency Identity Protocols Generation-2 UHF RFID Standard](#), release 2.1
 - **I** Definition: Width, Poly, Init, Residue (Table 6-11, p.33)
 - **I** Definition: RefIn, Residue (Annex F.2, p.143)
 - **I** Shift register circuit diagram (Figure F-2, p.144)
 - **I** Transmission order, implying RefIn and RefOut (Section 6.3.1.4, p.33)
 - **IV** 7 codewords (Table F-2, pp.144–5)
 - 0000E2F0
 - 08001111CCAE
 - 100011112222968F
 - 180011112222333378F6
 - 20001111222233334444C241
 - 280011112222333344445555A91
 - 30001111222233334444555566661835
- Philips Semiconductors (30 January 2004), [SL2 ICS11 Product Specification](#), revision 3.0 (courtesy of the Internet Archive)
 - **I** Definition: Width, Poly, Init, Residue (Section 10.2, p.26)
 - **I** Code: C (Section 10.3.2, pp.30–32)
 - **I** Worked example (as code trace) (Section 10.3.2, p.32)
- ETSI [EN 300 751](#) version 1.2.1 (January 2003)
 - **I** Definition: Width, Poly (Section 11.2.5, p.68)

- **IV** 1 codeword (Section 11.2.5, p.68)
 - 402141424387F5
- PVL Team (25 October 2008), CRC .NET control, version 14.0.0.0
 - **II** Implementation
- Hee-Bok Kang *et al.* (December 2008), [High Security FeRAM-Based EPC C1G2 UHF \(860 MHz-960 MHz\) Passive RFID Tag Chip](#), ETRI Journal, Volume 30 Number 6
 - **IV** 2 codewords (p.830)
 - 10100000000100000000000000000101000001010001
 - 3000319F60A356276E28F14FDC9C5DB4
- "Geert" (7 December 2007), Computer Interfacing Forum [topic 216](#) (courtesy of the Internet Archive)
 - **III** Quoted definition for GENIbus: Width, Poly, Init, XorOut
 - **IV** 2 codewords cited for GENIbus
 - 100120020C823E003982150064820900FA910A
 - 07200102C302101A901C
- Scott McLaughlin (24 September 2008), Computer Interfacing Forum [topic 907](#) (courtesy of the Internet Archive)
 - **III** Reported definition for TI Tag-It: full (except Check, Residue)
 - **IV** 4 codewords cited for TI Tag-It
 - 0000000001000000000101100011110010111
 - 0000000001010000000000000001100010110001101000000101101101000111001
 - 110000000101000000000000000110001011000110100000010000001000111110010
01111011010001011100000011000001
 - 000000010101000000000000000110001011000110100000011000001110000000101
010101010101100010011000111
- Unique effective solution of codeword set
- Created: 2 November 2007
- Updated: 3 January 2021

CRC-16/GSM

width=16 poly=0x1021 init=0x0000 refin=false refout=false xorout=0xffff check=0xce3c
residue=0x1d0f name="CRC-16/GSM"

- Class: attested
- ETSI [TS 100 909](#) version 8.9.0 (January 2005)
 - **I** Full mathematical description (Section 5.1.2.2, p.67 — Section 5.1.3.2, p.69 — Section 5.1.4.2, p.70)
- [ECMA standard ECMA-130](#), edition 2 (June 1996) — ISO/IEC 10149:1995
 - **I** Definition: Width, Poly, XorOut, RefOut (Section 22.3.6, p.23)
- Andreas Mueller *et al.* (15 May 2018), [cdrdao](#) 1.2.4, modules cdrdao-1.2.4/dao/SubChannel.cc, cdrdao-1.2.4/dao/PQSubChannel16.cc
 - **III** Code: C++
- Stanislav Brabec (26 January 2018), [CD-disassembler 0.1](#) (gzipped tar archive; courtesy of the Internet Archive), module cddisasm-0.1/cddisasm.c
 - **III** Code: C
- Stanislav Brabec (17 January 2014), [CD-disassembler](#) homepage (courtesy of the Internet Archive)
 - **IV** 2 codewords
 - 0101010000000000002005A28
 - 035D24E3580200010034D0F2
- Created: 17 February 2017
- Updated: 4 August 2023

CRC-16/IBM-3740

width=16 poly=0x1021 init=0xffff refin=false refout=false xorout=0x0000 check=0x29b1
residue=0x0000 name="CRC-16/IBM-3740"

- Class: attested
- Alias: **CRC-16/AUTOSAR**, **CRC-16/CCITT-FALSE**
- An algorithm commonly misidentified as CRC-CCITT. CRC-CCITT customarily refers to the LSB-first form of the algorithm in ITU-T Recommendation V.41 (see [CRC-16/KERMIT](#)); its MSB-first counterpart is [CRC-16/XMODEM](#).
- AUTOSAR (24 November 2022), [AUTOSAR Classic Platform](#) release R22-11, [Specification of CRC Routines](#)
 - I Comprehensive primer on CRC theory (Section 7.1, pp.19–22)
 - I All parameters (Section 7.2.2.1, p.26)
 - IV 7 codewords (Section 7.2.2.1, p.26)
 - 0000000084C0
 - F20183D374
 - 0FAA00552023
 - 00FF5511B8F9
 - 332255AABCCDDEEFF53F
 - 926B550745
 - FFFFFFFF1D0F
- Western Digital Corporation (May 1980), [FD 179X-02 datasheet](#) (courtesy of University of Kent Mirror Service)
 - I Definition: Width, Poly, Init (p.5)
- Floppy disc formats: IBM 3740 (FM, e.g. Acorn DFS), ISO/IEC 8860-2:1987 (DOS 720K), ISO/IEC 9529-2:1989 (DOS 1.4M)
 - II Implementation
- Lammert Bies (August 2011), [CRC calculator](#)
 - II Implementation
- PVL Team (25 October 2008), CRC .NET control, version 14.0.0.0
 - II Implementation
- Dr Ross N. Williams (19 August 1993), ["A Painless Guide to CRC Error Detection Algorithms"](#) (courtesy of the Internet Archive)
 - III All parameters (except Check, Residue)
- Berndt M. Gammel (29 October 2006), [Matpack 1.9.1](#) class MpCRC documentation
 - III All parameters (except Residue)
- Altera Corporation (April 1999), [crc MegaCore Function Data Sheet](#), version 2 (courtesy of the Internet Archive)
 - III All parameters (except Residue) (p.6)
- Scarybeasts (19 February 2021), [Beebjit](#) module [beebjit/ibm_disc_format.c](#)
 - III Code: C
 - IV 2 codewords
 - FE0000001F1D3
 - FB + E5 × 256 + A40C
- Unique effective solution of codeword set
- Created: 30 March 2005
- Updated: 3 August 2023

CRC-16/IBM-SDLC

width=16 poly=0x1021 init=0xffff refin=true refout=true xorout=0xffff check=0x906e
residue=0xf0b8 name="CRC-16/IBM-SDLC"

- Class: attested
- Alias: **CRC-16/ISO-HDLC**, **CRC-16/ISO-IEC-14443-3-B**, **CRC-16/X-25**, **CRC-B**, **X-25**
- HDLC is defined in ISO/IEC 13239. CRC_B is defined in ISO/IEC 14443-3.
- [ITU-T Recommendation T.30](#) (September 2005)
 - I Definition: Residue; full mathematical description (Section 5.3.7, p.78)
- [ITU-T Recommendation V.42](#) (March 2002)
 - I Definition: Residue; full mathematical description (Section 8.1.1.6.1, p.17)
- [ITU-T Recommendation X.25](#) (October 1996)
 - I Definition: Residue; full mathematical description (Section 2.2.7.4, p.9)
 - IV 4 codewords (Appendix I, p.145)
 - 033F5BEC
 - 01738357
 - 013FEBDF
 - 03733364
- [IETF RFC 1331](#) (May 1992)
 - I Code: C (Appendix B, pp.61–3)
- [IETF RFC 1662](#) (July 1994)
 - I Code: C (Appendix C.2, pp.19–21)
- William H. Press, Brian P. Flannery, Saul A. Teukolsky, William T. Vetterling (1992), *Numerical recipes in C: The art of scientific computing*, 2nd ed., Cambridge University Press, ISBN 0-521-43108-5
 - III All parameters (except Check) (p.898)
 - III Code: C (pp.900–1)
 - IV 2 codewords (p.898)
 - 54D9E4 (source: 54261B)
 - 4361744D6F757365393837363534333231910A (source: 4361744D6F7573653938373635343332316EF5)
- Berndt M. Gammel (29 October 2006), [Matpack 1.9.1](#) class MpCRC documentation
 - III All parameters (except Residue)
- iBiquity Digital Corporation (14 December 2016), [HD Radio™ Air Interface Design Description: Advanced Application Services Transport](#), rev. H (courtesy of National Radio Systems Committee)
 - IV 1 codeword (Figure 6-3, p.17)
 - 21005100004944330300000000004A544954320000000D000000416E616C6F6720426C756573545045310000000D0000004A2E20512E205075626C696354414C4200000012000000546865204C6F73742053657373696F6E73F527
- [ISO/IEC FCD 14443-3](#) (24 November 2008)
 - IV 3 codewords (Annex B.2, p.50)
 - 000000CCCC6
 - 0FAAFFFCDD1
 - 0A1234562CF6
- Unique effective solution of codeword set
- Created: 10 September 2005
- Updated: 4 August 2023

CRC-16/ISO-IEC-14443-3-A

width=16 poly=0x1021 init=0xc6c6 refin=true refout=true xorout=0x0000 check=0xbf05
residue=0x0000 name="CRC-16/ISO-IEC-14443-3-A"

- Class: attested
- Alias: **CRC-A**
- Used in contactless IC cards.
- [ISO/IEC FCD 14443-3](#) (24 November 2008)

- I Definition: Init, XorOut (Section 6.2.4, p.11)
- I Citation for rest of algorithm: ISO/IEC 13239 (see [CRC-16/IBM-SDLC](#))
- IV 2 codewords (Annex B.1, p.49)
 - 0000A01E
 - 123426CF
- Adam Laurie (31 August 2010), [crc16.c](#) (C source text; courtesy of the Internet Archive)
 - III Code: C
- Adam Laurie (31 August 2010), [comment on a Ridrix's Blog post](#)
 - III Code: C
 - IV 1 codeword
 - AABBCCDDEE0011227B09
- Flavio D. Garcia, *et al.* (October 2008), [Dismantling MIFARE Classic](#), Proceedings of the Thirteenth European Symposium on Research in Computer Security (ESORICS 2008)
 - IV 6 codewords (Figure 2, p.100; Section 4.1, p.102, referencing ISO/IEC 14443; Figure 6, p.105)
 - 9370C2A82DF4B3BAA3
 - 08B6DD
 - 6030764A
 - 500057CD
 - 937000000000009CD9
 - 6000F57B
- Flavio D. Garcia, *et al.* (October 2008), [Dismantling MIFARE Classic](#), slide presentation, ESORICS 2008 conference
 - IV 1 additional codeword (p.13)
 - 93701DFBE03335D355
- Unique effective solution of codeword set
- Created: 30 April 2011
- Updated: 3 August 2023

CRC-16/KERMIT

```
width=16 poly=0x1021 init=0x0000 refin=true refout=true xorout=0x0000 check=0x2189
residue=0x0000 name="CRC-16/KERMIT"
```

- Class: attested
- Alias: **CRC-16/BLUETOOTH, CRC-16/CCITT, CRC-16/CCITT-TRUE, CRC-16/V-41-LSB, CRC-CCITT, KERMIT**
- Used in Bluetooth error detection. Init=0x0000 is used in the Inquiry Response substate.
- Press *et al.* identify the CCITT algorithm with the one implemented in Kermit. V.41 is endianness-agnostic, referring only to bit sequences, but the CRC appears reflected when used with LSB-first modems. Ironically, the unreflected form is used in [CRC-16/XMODEM](#).
- For the algorithm often misidentified as CCITT, see [CRC-16/IBM-3740](#).
- [ITU-T Recommendation V.41](#) (November 1988)
 - I Definition: Residue; full mathematical description (Section 2, p.2)
 - I Shift register diagrams (Appendix I, p.9)
- Frank da Cruz (June 1986), [Kermit Protocol Manual, Sixth Edition](#) (plain text; courtesy of the Internet Archive)
 - I Full definition (except Check) (Section 6.3)
 - I Pseudocode (Section 6.3)
- Bluetooth SIG (31 January 2023), [Bluetooth Specification, Core Version 5.4](#)
 - I Definition: Init (Section 2.B.1.2.1, p.425; Section 2.B.6.5.1.4, pp.481–3 — Section 2.B.6.6.2, pp.490–3 — Section 2.B.7.1, p.497 — Section 2.B.8.4.3, pp.525–7)
 - I Definition: Width, Poly, RefIn, RefOut, XorOut (Section 2.B.7.1.2, pp.498–9)

- # CRC-16/LJ1200

width=16 poly=0x6f63 init=0x0000 refin=false refout=false xorout=0x0000 check=0xbdf4
residue=0x0000 name="CRC-16/LJ1200"

- Class: **third-party**
- Reference from named polynomial in Prof. Koopman's [CRC Polynomial Zoo](#).
- Used in the LoJack vehicle tracking system.
- Dennis Sheirer (24 January 2015), [SDRTrunk](#) module
[sdrtrunk/src/main/java/io/github/dsheirer/edac/CRCLJ.java](#)
 - **III** Code: Java
 - **III** Worked example (as generator matrix)
- Created: 20 February 2016
- Updated: 6 June 2018

CRC-16/M17

width=16 poly=0x5935 init=0xffff refin=false refout=false xorout=0x0000 check=0x772b
residue=0x0000 name="CRC-16/M17"

- Class: attested
- Contributed by Mark Adler.
- M17 Project (13 July 2023), [M17 protocol specification](#), version 1.0
 - **I** Definition: Width, Poly, Init, RefIn, RefOut, Check (Section 2.5.4, pp.24–5; Table 2.9, p.25)
 - **IV** 3 codewords (Table 2.9, p.25)
 - FFFF
 - 41206E
 - $(i) \times 256 + 1c31$
- Created: 7 May 2022
- Updated: 8 August 2024

CRC-16/MAXIM-DOW

width=16 poly=0x8005 init=0x0000 refin=true refout=true xorout=0xffff check=0x44c2
residue=0xb001 name="CRC-16/MAXIM-DOW"

- Class: attested
- Alias: **CRC-16/MAXIM**
- Maxim Integrated (8 August 2012), [DS1921G Datasheet](#)
 - **I** Definition: Width, Poly, Init, XorOut (p.32)
 - **I** Shift register diagram (p.34)
- Maxim Integrated (22 August 2012), [Application Note 27](#) (PDF, [HTML](#))
 - **I** Definition: Init (p.10)
 - **I** Shift register diagram (p.11)
 - **I** Code: 8051 assembler (pp.12–16)
 - **I** Worked example (p.15)
- Created: 26 April 2009
- Updated: 28 December 2019

CRC-16/MCRF4XX

width=16 poly=0x1021 init=0xffff refin=true refout=true xorout=0x0000 check=0x6f91
residue=0x0000 name="CRC-16/MCRF4XX"

- Class: attested

- Youbok Lee, PhD, Microchip Technology Inc. (16 July 2001), ["CRC Algorithm for MCRF45X Read/Write Device"](#)
 - I Definition: Width, Poly (reverse form) (p.1)
 - I Shift register diagram (p.1)
 - I Flowchart (p.2)
 - I Worked example (p.3)
 - I Code: C (pp.4–5)
- Piers Desrochers, PICList MassMind, ["A quick guide to CRC"](#) (courtesy of the Internet Archive)
 - III Description
 - III Worked example
- William H. Press, Brian P. Flannery, Saul A. Teukolsky, William T. Vetterling (1992), *Numerical recipes in C: The art of scientific computing*, 2nd ed., Cambridge University Press, ISBN 0-521-43108-5
 - III All parameters (except Check) (p.898)
 - III Code: C (pp.900–1)
 - IV 2 codewords (p.898)
 - 54261B
 - 4361744D6F7573653938373635343332316EF5
- "kilobravo" (16 June 2008), Computer Interfacing Forum [topic 578](#) (courtesy of the Internet Archive)
 - IV 2 codewords
 - 3A7164D9
 - 3A916A3E
- Unique effective solution of codeword set
- Created: 18 January 2008
- Updated: 4 August 2023

CRC-16/MODBUS

width=16 poly=0x8005 init=0xffff refin=true refout=true xorout=0x0000 check=0x4b37
residue=0x0000 name="CRC-16/MODBUS"

- Class: attested
- Alias: **MODBUS**
- CRC presented low byte first.
- MODICON Inc. (June 1996), [Modbus Protocol Reference Guide](#), Rev. J
 - I Algorithm (pp.112–3)
 - I Code: C (pp.113–5)
- Lammert Bies (August 2011), [CRC calculator](#)
 - II Implementation
- Ondřej Karas (28 February 2006), CRC calculator
 - II Implementation
- Panu-Kristian Poiksalo (28 February 2003), Control.com [forum post](#)
 - III Code: ObjectPascal
- Created: 27 February 2007
- Updated: 2 January 2021

CRC-16/NRSC-5

width=16 poly=0x080b init=0xffff refin=true refout=true xorout=0x0000 check=0xa066
residue=0x0000 name="CRC-16/NRSC-5"

- Class: attested

- Implemented in standard NRSC-5-D as a 16-bit CRC truncated (punctured) to 12 bits. The upper 4 bits, the last to leave the shift register, are discarded.
- Init = 0xFFFF implements the algorithm prescribed by iBiquity on 80-bit PDUs. Substitute Init = 0x39C8 if skipping 8 leading zeroes, Init = 0x849D if skipping all 12 leading zeroes and processing only the 68 message bits. In any case Init = 0x0000, XorOut = 0x4955 is equivalent.
- iBiquity Digital Corporation (23 August 2011), [HD Radio™ Air Interface Design Description: Station Information Service Transport](#), rev. J (courtesy of National Radio Systems Committee)
 - I Definition: Poly (mantissa only); method (Section 4.10, p.38)
 - Incorporated by normative reference into National Radio Systems Committee standard [NRSC-5-D](#).
- Andrew Wesie (9 June 2017), [NRSC-5 receiver for rtl-sdr](#), file [nrsc5/support/sample.xz](#) (XZ compressed raw radio recording, 22 MB)
 - II Recording confirming implementation with 28 codewords (4 bits extrapolated; selection)
 - 000048BBABA0329A0A0A2671 (source: 0000101000001010100110100011001010100000101010111011101101001000000000100100110)
 - 0030C8D3D37379D91E0A7B40 (source: 00001010000011110110110010111100101110011110100111101001111001100100000011000001111011)
 - 0010C097A920F31B240A2D1D (source: 000010100010010000001101111111001100100000101010011001011111000000001110100101101)
 - 00200882E861D03C3C0AEB71 (source: 0000101000111100000111100110100000110000111101000100000100000100000100000111101011)
 - 00100000DC69FABC4E0AB6CD (source: 0000101001001110101111001111101001101001110111000000000000000000001110110110110)
 - 000015004000809E54456AC7 (source: 01000101010101001001111101000000000000000001010000000000000001101101010)
 - 00200008FA7E849E54454B82 (source: 010001010101010010011111010000100011111101111101000001000000000000001000100101011)
 - 001000744280849E54457C5C (source: 010001010101010010011111010000100100000000010000100111010000000000001110001111100)
- Andrew Wesie, Clayton Smith *et al.* (17 April 2018), [NRSC-5 receiver for rtl-sdr](#), module [nrsc5/src/pids.c](#)
 - III Code: C
- Created: 16 November 2018
- Updated: 20 March 2019

CRC-16/OPENSAFETY-A

```
width=16 poly=0x5935 init=0x0000 refin=false refout=false xorout=0x0000 check=0x5d38
residue=0x0000 name="CRC-16/OPENSAFETY-A"
```

- Class: attested
- The single codeword is supported by the codewords confirming [CRC-8/OPENSAFETY](#), defined identically apart from Width, Poly in the same standard.
- Ethernet POWERLINK Standardisation Group (18 October 2018), [OpenSAFETY Safety Profile Specification Working Draft Proposal](#), version 1.5.2
 - I Full description (Section 5.1.2.6, pp.43–4)
 - I Error detection capability (Section 8.1, p.182)
 - I Code: C (Appendix 1, pp.196–9)
 - IV 1 codeword (Appendix 1, l.3525, pp.196–7)
 - 23C8083411223344556677880374

- Prof. Philip Koopman, Tridib Chakravarty (June 2004), [Cyclic Redundancy Code \(CRC\) Polynomial Selection for Embedded Networks](#)
 - III Assessment of polynomial performance (as 0xAC9A or C₃)
- Created: 22 July 2016
- Updated: 7 May 2022

CRC-16/OPENSAFETY-B

width=16 poly=0x755b init=0x0000 refin=false refout=false xorout=0x0000 check=0x20fe
residue=0x0000 name="CRC-16/OPENSAFETY-B"

- Class: attested
- The single codeword is supported by the codewords confirming [CRC-8/OPENSAFETY](#), defined identically apart from Width, Poly in the same standard.
- Ethernet POWERLINK Standardisation Group (18 October 2018), [OpenSAFETY Safety Profile Specification Working Draft Proposal](#), version 1.5.2
 - I Full description (Section 5.1.2.6, pp.43–4)
 - I Error detection capability (Section 8.1, p.182)
 - I Code: C (Appendix 1, pp.196–9)
 - IV 1 codeword (Appendix 1, I.3530, pp.196–7)
 - 22C812563011223344556677887031
- Prof. Philip Koopman, Tridib Chakravarty (June 2004), [Cyclic Redundancy Code \(CRC\) Polynomial Selection for Embedded Networks](#)
 - III Assessment of polynomial performance (as 0xBAAD)
- Created: 22 July 2016
- Updated: 7 May 2022

CRC-16/PROFIBUS

width=16 poly=0x1dcf init=0xffff refin=false refout=false xorout=0xffff check=0xa819
residue=0xe394 name="CRC-16/PROFIBUS"

- Class: attested
- Alias: **CRC-16/IEC-61158-2**
- PROFIBUS International (March 1998), [PROFIBUS Specification, Normative Parts](#), Edition 1.0, Part 9 (courtesy of the Internet Archive)
 - I Transmission order, implying RefIn and RefOut (Part 9 Section 8.5.1, p.902)
 - I Definition: Width, Poly, Init, Residue (Part 9 Section 8.8.4, pp.905–7)
 - I Properties of polynomial (Part 9 Section 8.8.4, p.906)
- "Chetanya" (17 November 2011), Computer Interfacing Forum [topic 1752](#) (courtesy of the Internet Archive)
 - III Definition: Width, Poly, Init
 - IV 5 codewords
 - 34AF21
 - 011057E0
 - 321000E37E82
 - 331000E32A0F
 - 261101000080A109739
- Unique effective solution of codeword set
- Created: 5 July 2016
- Updated: 7 May 2022

CRC-16/RIELLO

```
width=16 poly=0x1021 init=0xb2aa refin=true refout=true xorout=0x0000 check=0x63d0
residue=0x0000 name="CRC-16/RIELLO"
```

- Class: **third-party**
- Reported for a Riello Dialog UPS.
- "Snamprogetti" (17 April 2009), Computer Interfacing Forum [topic 1305](#) (courtesy of the Internet Archive)
 - **III** Definition: Width, Poly, Init
 - **III** Algorithm reported to be [CRC-CCITT](#), implying RefIn and RefOut.
 - **IV** 1 codeword
 - 2022525330308790
- Created: 18 April 2009
- Updated: 28 December 2019

CRC-16/SPI-FUJITSU

```
width=16 poly=0x1021 init=0x1d0f refin=false refout=false xorout=0x0000 check=0xe5cc  
residue=0x0000 name="CRC-16/SPI-FUJITSU"
```

- Class: attested
- Alias: **CRC-16/AUG-CCITT**
- Init value is equivalent to an augment of 0xFFFF prepended to the message.
- Fujitsu Semiconductor (10 October 2007), [FlexRay ASSP MB88121B User's Manual](#) (courtesy of the Internet Archive)
 - I Definition: Width, Poly, Init (Section 2.5.1, p.153)
- Lammert Bies (August 2011), [CRC calculator](#)
 - II Implementation
- PVL Team (25 October 2008), CRC .NET control, version 14.0.0.0
 - II Implementation
- Berndt M. Gammel (29 October 2006), [Matpack 1.9.1](#) class MpCRC documentation
 - III All parameters (except Residue)
- Created: 30 October 2007
- Updated: 4 August 2023

CRC-16/T10-DIF

```
width=16 poly=0x8bb7 init=0x0000 refin=false refout=false xorout=0x0000 check=0xd0db
residue=0x0000 name="CRC-16/T10-DIF"
```

- [illegible]

- FFFFEFD FCBFAF9F8F7F6F5F4F3F2F1F0E FEEDECEBEAE9E8E7E6E5E4E3E2E1E0A0B7
- Ralph Weber, John Lohmeyer, INCITS Technical Committee T10 (20 August 2003), [Minutes of Data Integrity Study Group - Aug 19-20, 2003](#)
 - I Definition: Init, XorOut (Item 4.6, p.3)
 - I Acceptance of previous two documents (Item 4.7, p.3)
- Pat Thaler, INCITS Technical Committee T10 (28 August 2003), [16-bit CRC polynomial selection](#)
 - I Properties of polynomial
- Martin K. Petersen, Oracle Corporation (2007); Tim Chen, Intel Corporation (2013), Linux module [crypto/crct10dif_common.c](#)
 - III Code: C
- Created: 29 September 2009
- Updated: 2 January 2021

CRC-16/TELEDISK

width=16 poly=0xa097 init=0x0000 refin=false refout=false xorout=0x0000 check=0x0fb3
residue=0x0000 name="CRC-16/TELEDISK"

- Class: confirmed
- Used in the Teledisk disc archive format. DECnet and [CRCK](#) allegedly use a Sick-type algorithm but with this polynomial.
- Will Kranz (8 December 2002), [wteledsk 1.01](#)
 - II Implementation
- Will Kranz (8 December 2002), [wteledsk 1.01](#) module [src/tdcrc.c](#)
 - III Code: C
- Created: 10 December 2009
- Updated: 4 August 2023

CRC-16/TMS37157

width=16 poly=0x1021 init=0x89ec refin=true refout=true xorout=0x0000 check=0x26b1
residue=0x0000 name="CRC-16/TMS37157"

- Class: attested
- Texas Instruments, Inc. (November 2009), [TMS37157 datasheet](#)
 - I Full definition (except Check) (pp.39–40)
 - I Shift register diagram (Figure 51, p.39)
 - I Flowchart (Figure 52, p.40)
- "harper", "jacknad" (November 2010–January 2011), StackOverflow [submitted question and answers](#)
 - III Definition: Init (reverse form) ("harper")
 - III Code: C ("jacknad")
 - IV 3 codewords ("harper")
 - 0225A6
 - 024000000000A060E7
 - 024100000000A46FA5
- "Petri" (4 May 2010), Texas Instruments, Inc. E2E™ Community [forum post](#)
 - IV 1 codeword
 - 01AAAAAAAAADC25AB
- Unique effective solution of codeword set
- Created: 17 April 2011
- Updated: 28 December 2019

CRC-16/UMTS

width=16 poly=0x8005 init=0x0000 refin=false refout=false xorout=0x0000 check=0xfee8
residue=0x0000 name="CRC-16/UMTS"

- Class: attested
- Alias: **CRC-16/BUYPASS**, **CRC-16/VERIFONE**
- Reported for the multi-threaded portion of the Bypass transaction processing network.
- Verifone, Inc. (May 1995), [TCLOAD Reference Manual](#) (courtesy of the Internet Archive)
 - I Definition: Poly. CRC byte order, implying RefIn and RefOut (Section 4, p.4-1)
- 3rd Generation Partnership Project (3GPP) [TS 25.427](#) version 17.0.0 (April 2022) (zipped MS Word document) — ETSI [TS 125 427](#) version 17.0.0 (April 2022)
 - I Definition: Width, Poly, Init, XorOut (Section 7.2.1, p.40)
 - I Definition: RefIn, RefOut (Section 6.1.1, p.20; Section 7.2.1.1, p.40)
- PVL Team (25 October 2008), CRC .NET control, version 14.0.0.0
 - II Implementation
- Emil Lenchak, Texas Instruments, Inc. (June 2018), [CRC Implementation With MSP430](#)
 - III All parameters (except Residue) (Section 4.6, p.6)
- Thomas Schmidt, Microchip Technology Inc. (20 May 2000), ["CRC Generating and Checking"](#)
 - III Description (p.4)
 - III Shift register diagram (Figure 1, p.3)
 - III Code: 16C54B assembler (Appendix A, pp.7–11; Appendix B, pp.12–22)
- Michael Niedermayer, Aurélien Jacobs *et al.* (27 February 2023), [FFmpeg 6.0](#) (XZ compressed tarfile), module [ffmpeg-6.0/libavutil/tests/crc.c](#)
 - IV 1 codeword
 - $(i+i*i) \times 1999 + 1\text{FBB}$
- "Joshua" (31 May 2008), Computer Interfacing Forum [topic 530](#) (courtesy of the Internet Archive)
 - IV 2 codewords
 - 0384901B56
 - 03848400001230314131333030323031333030311C39303062BF
- Michael Junker (26 October 2009), Computer Interfacing Forum [topic 1432](#) (courtesy of the Internet Archive)
 - IV 7 codewords (7-bit payload characters)
 - 01100110011101011000001100000110011001110101100001101000111010100 (source: 331D3030331D30 + 53716)
 - 01101000011101011000001100000110011001110101100001111000111001001 (source: 341D3030331D30 + 61897)
 - 0110101001110101100000110000011001100111010110000001000111001111 (source: 351D3030331D30 + 04559)
 - 01101100011101011000001100000110011001110101100001011000111000000 (source: 361D3030331D30 + 45504)
 - 01100110011101011010001100000110000001110101100000101000111000110 (source: 331D3430301D30 + 20934)
 - 01100110011101011000001100000111001001110101100001101000111110110 (source: 331D3030391D30 + 53750)
 - 0110001001110101100000110001011000000111010110000001001101010001 (source: 311D3031301D30 + 04945)
- Unique effective solution of codeword set
- Created: 4 June 2008
- Updated: 8 August 2024

CRC-16/USB

width=16 poly=0x8005 init=0xffff refin=true refout=true xorout=0xffff check=0xb4c8
residue=0xb001 name="CRC-16/USB"

- Class: confirmed
- CRC appended low byte first.
- Cypress Semiconductor Corporation (26 July 2001), [SL811HS/SL811HST Application Notes](#), Revision 1.21 (courtesy of Stardot.org.uk Forums)
 - II Transcript confirming implementation with 3 codewords (pp.25,30)
 - 1BED534B3E6D7F7CD7CC (source: CRC = EB33)
 - 55534243A864AB870800000080000A250000000000000000000000000000C30E (source: CRC = C370)
 - 8006000100004000DD94 (source: CRC = BB29)
- Anonymous (10 July 1997), ["Cyclic Redundancy Checks in USB"](#) (Draft) (courtesy of the Internet Archive)
 - III Definition: Width, Poly, Init, XorOut (p.2)
 - III Definition: Residue (p.3)
 - III Code: Perl (p.4)
 - IV 2 codewords (p.6)
 - 00010203EF7A
 - 234567890E1C
- Unique effective solution of codeword set
- Created: 1 December 2007
- Updated: 12 July 2024

CRC-16/XMODEM

width=16 poly=0x1021 init=0x0000 refin=false refout=false xorout=0x0000 check=0x31c3
residue=0x0000 name="CRC-16/XMODEM"

- Class: attested
- Alias: **CRC-16/ACORN, CRC-16/LTE, CRC-16/V-41-MSB, XMODEM, ZMODEM**
- The MSB-first form of the V.41 algorithm. For the LSB-first form see [CRC-16/KERMIT](#). CRC presented high byte first.
- Used in the MultiMediaCard interface. In XMODEM and Acorn MOS the message bits are processed out of transmission order, compromising the guarantees on burst error detection.
- [ITU-T Recommendation V.41](#) (November 1988)
 - I Definition: Residue; full mathematical description (Section 2, p.2)
 - I Shift register diagrams (Appendix I, p.9)
- 3rd Generation Partnership Project (3GPP) [TS 36.212](#) version 17.1.0 (March 2022) (zipped MS Word document) — ETSI [TS 136 212](#) version 17.1.0 (April 2022)
 - I Definition: Width, Poly, Init, XorOut, Residue (Section 5.1.1, pp.10–11)
 - I Attachment relation, defining RefIn ^ RefOut (Section 5.1.1, p.11)
- 3rd Generation Partnership Project (3GPP) [TS 36.321](#) version 17.5.0 (June 2023) (zipped MS Word document) — ETSI [TS 136 321](#) version 17.5.0 (July 2023)
 - Referenced from TS 136 212 section 5.2.2.1, p.26
 - I Definition: RefIn, RefOut (Section 6.1.1, p.90)
- JEDEC Standard [JESD84-A441](#) (March 2010) (registration required)
 - I Full definition (Section 10.2, pp.157–8)
 - I Shift register diagram (Figure 54, p.159)
- Acorn Computers Ltd (October 1984), [BBC Microcomputer User Guide](#) (courtesy of the Internet Archive)
 - I Pseudocode (Chapter 35, p.369)
- Ward Christensen, Keith Petersen *et al.* (8 June 1982), XMODEM 5.0

- II Implementation (attributed to Paul Hansknecht)
- Acorn Computers Ltd (1981), Acorn MOS 1.20 (BBC Micro cassette format)
 - II Implementation
- Lammert Bies (August 2011), [CRC calculator](#)
 - II Implementation
- PVL Team (25 October 2008), CRC .NET control, version 14.0.0.0
 - II Implementation
- Berndt M. Gammel (29 October 2006), [Matpack 1.9.1](#) class MpCRC documentation
 - III All parameters (except Residue)
- Altera Corporation (April 1999), [crc MegaCore Function Data Sheet](#), version 2 (courtesy of the Internet Archive)
 - III All parameters (except Residue) cited for ZMODEM (p.6)
- William H. Press, Brian P. Flannery, Saul A. Teukolsky, William T. Vetterling (1992), *Numerical recipes in C: The art of scientific computing*, 2nd ed., Cambridge University Press, ISBN 0-521-43108-5
 - III All parameters (except Check) (p.898)
 - III Code: C (pp.900–1)
 - IV 2 codewords (p.898)
 - 541A71
 - 4361744D6F757365393837363534333231E556
- Created: 30 March 2005
- Updated: 5 August 2023

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